MT8002 Elements of Set Theory
Chapter 7. Ordered Sets
Section 7.4. Well-Ordered Sets

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Well-Ordered Sets
Some Properties of Well-Ordered Sets
Comparing Well-Ordered Sets
Initial Segments and Proof by Induction
References

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Well-Ordered Sets

Definition (p. 188)

Let $(X, <)$ be a linearly ordered set. The relation $<$ is a **well-order** (or the set X is **well-ordered** by the relation $<$) if every non-empty subset A of X has a least element, i.e., there is some element $a \in A$ such that $a \leq x$, for all $x \in A$.

Well-Ordered Sets

Definition (p. 188)

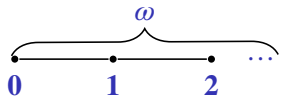
Let $(X, <)$ be a linearly ordered set. The relation $<$ is a **well-order** (or the set X is **well-ordered** by the relation $<$) if every non-empty subset A of X has a least element, i.e., there is some element $a \in A$ such that $a \leq x$, for all $x \in A$.[†]

[†]Note that the terms 'well-order' relation and 'well-ordered' set are not standard in the literature. For example, Enderton (1977) uses 'well ordering' for relations and 'well ordered' for sets and Button (2021) uses 'well-ordering' for both relations and sets.

Well-Ordered Sets

Example

The canonical example of a well-ordered set is ω, i.e., $(\mathbb{N}, \in)$,



because every non-empty subset of ω has a least element (Theorem 3.9).

Well-Ordered Sets

Example (informal)

The closed interval $[0, 1]$ of $\mathbb{R}$ with the usual order is not well-ordered because the interval $(0, 1) \subseteq [0, 1]$ has no least element.

Well-Ordered Sets

Example (informal)

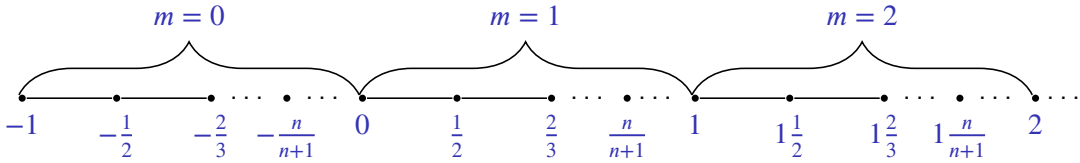
The closed interval $[0, 1]$ of $\mathbb{R}$ with the usual order is not well-ordered because the interval $(0, 1) \subseteq [0, 1]$ has no least element.

Let $a, b \in \mathbb{R}$, with $a < b$. In general, any subset of $\mathbb{R}$ containing an open interval (a, b) is not well-ordered.

Well-Ordered Sets

Example (informal)

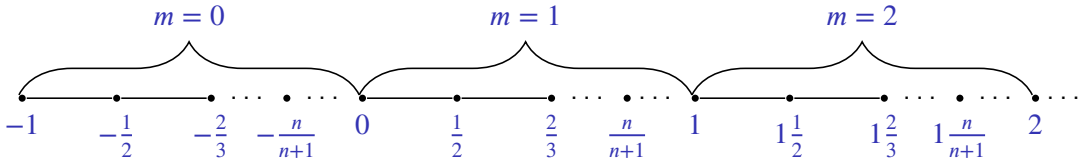
The subset $X = \left\{ m - \frac{1}{n+1} : m, n \in \mathbb{N} \right\}$ of $\mathbb{Q}$ with the usual order is well-ordered.



Well-Ordered Sets

Example (informal)

The subset $X = \left\{ m - \frac{1}{n+1} : m, n \in \mathbb{N} \right\}$ of $\mathbb{Q}$ with the usual order is well-ordered.



Note that

$$m - \frac{1}{n+1} < m' - \frac{1}{n'+1} \text{ iff } \begin{cases} m < m', \text{ or} \\ m = m' \text{ and } n < n'. \end{cases}$$

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Example (continuation)

Let A a non-empty subset of X.

"*The least element of A has to have the least m available among the elements of A and the smallest n for this m.*" *(p. 189)*

We find the least element of A using the well-order property of natural numbers.

Let $B = \left\{ m \in \mathbb{N} : m - \frac{1}{n+1} \in A, \text{ for some } n \in \mathbb{N} \right\}$ and let m_0 the least element of B.

Let $C = \left\{ n \in \mathbb{N} : m_0 - \frac{1}{n+1} \in A \right\}$ and let n_0 the least element of C.

Then $m_0 - \frac{1}{n_0+1}$ is the least element of A.

Outline

Well-Ordered Sets
Some Properties of Well-Ordered Sets
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Some Properties of Well-Ordered Sets

Theorem

Let $(X, <)$ be a well-ordered set and let $A \subseteq X$. Then the set A is well-ordered by the restriction of relation $<$ to A, i.e., $(A, <|_{A \times A})$ is a well-ordered set (Exercise 7.44).

Some Properties of Well-Ordered Sets

Example

The set $\mathbf{5} \doteq \{\mathbf{0}, \mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}\}$ is well-ordered by the relation $\in$.

Some Properties of Well-Ordered Sets

Let X be a set. Recall that an infinite descending $\in$-chain is a sequence $\langle x_n \rangle$ such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

Some Properties of Well-Ordered Sets

Let X be a set. Recall that an infinite descending $\in$-chain is a sequence $\langle x_n \rangle$ such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

Definition (p. 190)

Let $(X, <)$ be an ordered set. An **infinite descending $<$-chain** is a sequence $\langle x_n \rangle$ such that $x_{n+1} < x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} < x_n < \cdots < x_2 < x_1 < x_0.$$

Some Properties of Well-Ordered Sets

Theorem

Let $(X, <)$ be a well-ordered set and suppose that X contains an infinite descending $<$-chain. Then X is not well-ordered by $<$ (Exercise 7.45).

Some Properties of Well-Ordered Sets

Example (informal)

The set $\mathbb{Z}$ with the usual order $<$ is not a well-ordered set because the subset of negative integers forms an infinite descending $<$-chain.

$$\cdots < -3 < -2 < -1$$

Some Properties of Well-Ordered Sets

Theorem

Let X and Y be ordered sets and let $f : X \longrightarrow Y$ be an order-isomorphism. Then if X is well-ordered then so is Y (Exercise 7.46).

Some Properties of Well-Ordered Sets

Example

The set ω^2 is well-ordered because there is an order-isomorphism between it and the subset $\left\{m - \frac{1}{n+1} : m, n \in \mathbb{N}\right\}$ of $\mathbb{Q}$ with the usual order.

Some Properties of Well-Ordered Sets

Exercise 7.47

Suppose that X and Y are well-ordered sets. Show that the ordered sets $X + Y$ and $X \times Y$ are both well-ordered.

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Comparing Well-Ordered Sets

Comparison of ordered sets

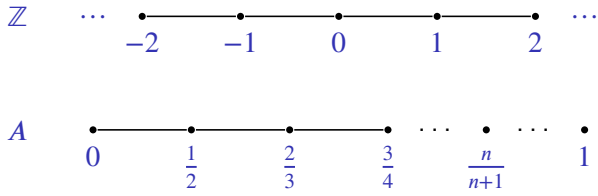
Two ordered sets are **comparable** if either they are order-isomorphic or one can be order-embedded into the other.

Any two well-ordered sets are comparable, but, two linearly ordered sets are not necessarily comparable.

Comparing Well-Ordered Sets

Example (informal)

The sets $\mathbb{Z}$ and $A = \left\{1 - \frac{1}{n+1} : n \in \mathbb{N}\right\} \cup \{1\}$, both with the usual order $<$, are incomparable (Exercise 7.50).



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Comparing Well-Ordered Sets

Example (continuation)

(a) $\mathbb{Z} \ncong A$ because 1 is a point limit of A but $\mathbb{Z}$ has not limit points.

(b) There is not an order-embedding $f : \mathbb{Z} \longrightarrow A$ because $\mathbb{Z}$ contains an infinite descending $<$-chain but A is a well-ordered set, i.e., A does not contain an infinite descending $<$-chain. Therefore we cannot order-embedding $\mathbb{Z}$ into A.

(c) There is not an order-embedding $f : A \longrightarrow \mathbb{Z}$ because "*whichever integers we chose for $f(0)$ and $f(1)$, there would only be finitely many possible images between them for the infinitely many $1 - \frac{1}{n+1}$, $n > 0$. So we could not define f to be both one-one and order-preserving.*" *(p. 191)*

Comparing Well-Ordered Sets

Comparison of well-ordered sets

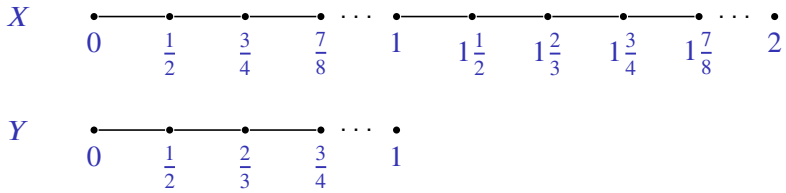
"*The basic idea is to pair off the least elements of X and Y, then their next least elements, and so on. If the process has worked for all the elements of X less than some limit point x of X, it turns out that the least available element of Y to be paired up with x also has to be a limit point. And if in this process we run out of elements of both X and Y at the same stage, we have an order-isomorphism; while if we run out of elements of one of the sets before we've used up all the elements of the other, then we have an order-embedding of one set into the other.*" *(p. 192)*

Comparing Well-Ordered Sets

Example (informal)

Let X and Y be well-ordered sets, subsets of $\mathbb{Q}$, with the usual order $<$, defined by

$$X = \left\{1 - \frac{1}{2^n} : n \in \mathbb{N}\right\} \cup \left\{2 - \frac{1}{2^n} : n \in \mathbb{N}\right\} \cup \{2\},$$
$$Y = \left\{1 - \frac{1}{n+1} : n \in \mathbb{N}\right\} \cup \{1\}.$$



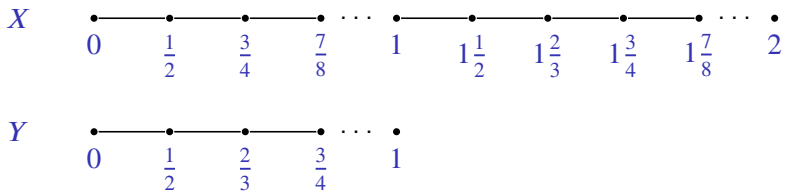
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Comparing Well-Ordered Sets

Example (informal)

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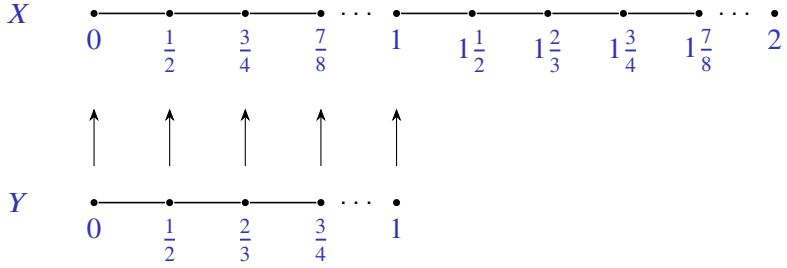


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[†]See the corrections to the textbook in the course homepage.

Comparing Well-Ordered Sets

Example (continuation)



The set Y runs out of elements before the set X, giving an order-embedding

$$\begin{aligned} f : Y &\longrightarrow X \\ 1 - \frac{1}{n+1} &\longmapsto 1 - \frac{1}{2^n}, \quad n \in \mathbb{N}, \\ 1 &\longmapsto 1. \end{aligned}$$

Comparing Well-Ordered Sets

Definition (p. 193)

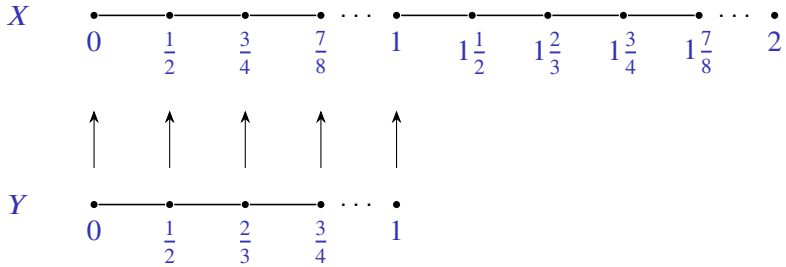
Let $(X, <)$ be a well-ordered and let $a \in X$. The **initial segment of X determined by a**, denoted $\mathrm{Seg}_X(a)$, is the set defined by

$$\mathrm{Seg}_X(a) := \{x \in X : x < a\}.$$

Comparing Well-Ordered Sets

Example (informal)

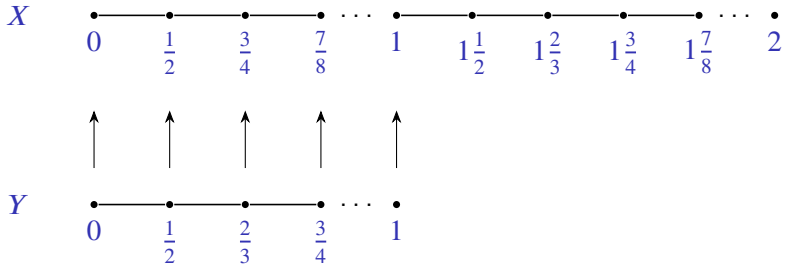
The order-embedding from the previous example maps the set Y onto the initial segment $\mathrm{Seg}_X(1\frac{1}{2})$ of the set X.



Comparing Well-Ordered Sets

Example (informal)

The order-embedding from the previous example maps the set Y onto the initial segment $\mathrm{Seg}_X(1\frac{1}{2})$ of the set X.†



†See the corrections to the textbook in the course homepage.

Comparing Well-Ordered Sets

Example

The initial segment of ω determined by $\mathbf{3}$ is the set

$$\begin{aligned}\mathrm{Seg}_{\omega}(\mathbf{3}) &\doteq \{\mathbf{n} \in \omega : \mathbf{n} \in \mathbf{3}\} \\ &\doteq \{\mathbf{0}, \mathbf{1}, \mathbf{2}\} \\ &\doteq \mathbf{3}.\end{aligned}$$

Comparing Well-Ordered Sets

Theorem 7.5

Let X and Y be well-ordered sets. Then exactly one of the following holds:

(a) the set X is order-isomorphic to an initial segment of the set Y,

(b) the set X is order-isomorphic to the set Y,

(c) the set Y is order-isomorphic to an initial segment of the set X.

Comparing Well-Ordered Sets

Theorem 7.5

Let X and Y be well-ordered sets. Then exactly one of the following holds:

(a) the set X is order-isomorphic to an initial segment of the set Y,

(b) the set X is order-isomorphic to the set Y,

(c) the set Y is order-isomorphic to an initial segment of the set X.

"*Theorem 7.5 is a very strong result: any two well-orders are essentially the same [i.e., order-isomorphic], or one is just the other with some extra elements tacked on at the end. It also suggests that one can organise well-ordered sets in terms of their 'length' as orders.*" *(p. 194)*

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Initial Segments and Proof by Induction

Theorem 3.5 (Proof by induction on $\mathbb{N}$)

For all subsets $A \subseteq \mathbb{N}$ such that:

(a) Base step: $\mathbf{0} \in A$.

(b) Inductive step: if $\mathbf{n} \in A$ then $\mathbf{n}^+ \in A$.

Then $A \doteq \mathbb{N}$.

Initial Segments and Proof by Induction

Theorem 3.5 (Proof by induction on $\mathbb{N}$)

For all subsets $A \subseteq \mathbb{N}$ such that:

(a) Base step: $\mathbf{0} \in A$.

(b) Inductive step: if $\mathbf{n} \in A$ then $\mathbf{n}^+ \in A$.

Then $A \doteq \mathbb{N}$.

Theorem 7.6 (Principle of transfinite induction)

Let X be a well-ordered set. For all subsets $A \subseteq X$ such that:

(a) Base step: the least element of X is in A.

(b) Inductive step: for all other $x \in X$, if $\mathrm{Seg}_X(x) \subseteq A$ then $x \in A$.

Then $A \doteq X$.

Technical remark

Note that the textbook excludes the least element of the set X from the inductive step of Theorem 7.6 because this element triviality satisfies the condition.

Initial Segments and Proof by Induction

Theorem 7.7

Let $(X, <)$ be a well-ordered set and let $f : X \longrightarrow X$ be an order-embedding of X into itself. Then $x \leq f(x)$, for all $x \in X$.

Initial Segments and Proof by Induction

Theorem 7.8

Let $(X, <)$ be a well-ordered set and let $f : X \longrightarrow X$ be an order-embedding of X into itself. Then f is the identity function.

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References

Tim Button (2021). Set Theory. An Open Introduction. Fall 2021. Revision 2138a50 (2026-07-11). Independently published. URL: https://builds.openlogicproject.org/courses/set-theory/ (cit. on p. 6).

Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 6).

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).